

AWP Construction Progress Dashboard

Construction Automation User Guide

Portfolio Project Demonstrating Construction Automation, AWP Progress Tracking, SQL Server Automation, and Power BI Field Reporting for EPC Projects. - Ikenna Ogenyi | github.com/OgenyiND/bechtel-awp-dashboard | July 2026

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1. Who should use this guide?

This guide is intended for construction engineers, project controls personnel, field supervisors, and construction automation technicians using the AWP Construction Progress Dashboard.

2. System overview

The AWP Construction Progress Dashboard is a simulated EPC field reporting system built to mirror Bechtel's Construction Automation workflows. It tracks daily construction progress across five disciplines and twenty contractors, flags data quality issues, and publishes KPI dashboards for field supervisors and project controls teams.

Component	Technology	Purpose
Database	SQL Server 2022	Stores contractors, work packages, daily progress
Data generation	Python (Faker)	Generates 1,420 rows of realistic EPC data
Validation queries	SQL (4 files)	Detects bad data, tracks progress, S-curve
Dashboard	Power BI Desktop	3-page field reporting dashboard
Daily automation	Python script	Inserts new records, exports flagged CSV report

Database tables

The BechtelAWP database contains three tables that simulate an active EPC project:

- **contractors** — 20 contractors with name, discipline, and compliance status
- **work_packages** — 600 IWPs with planned/actual quantities, dates, and status
- **daily_progress** — 800+ daily field records with qty installed and quality flags

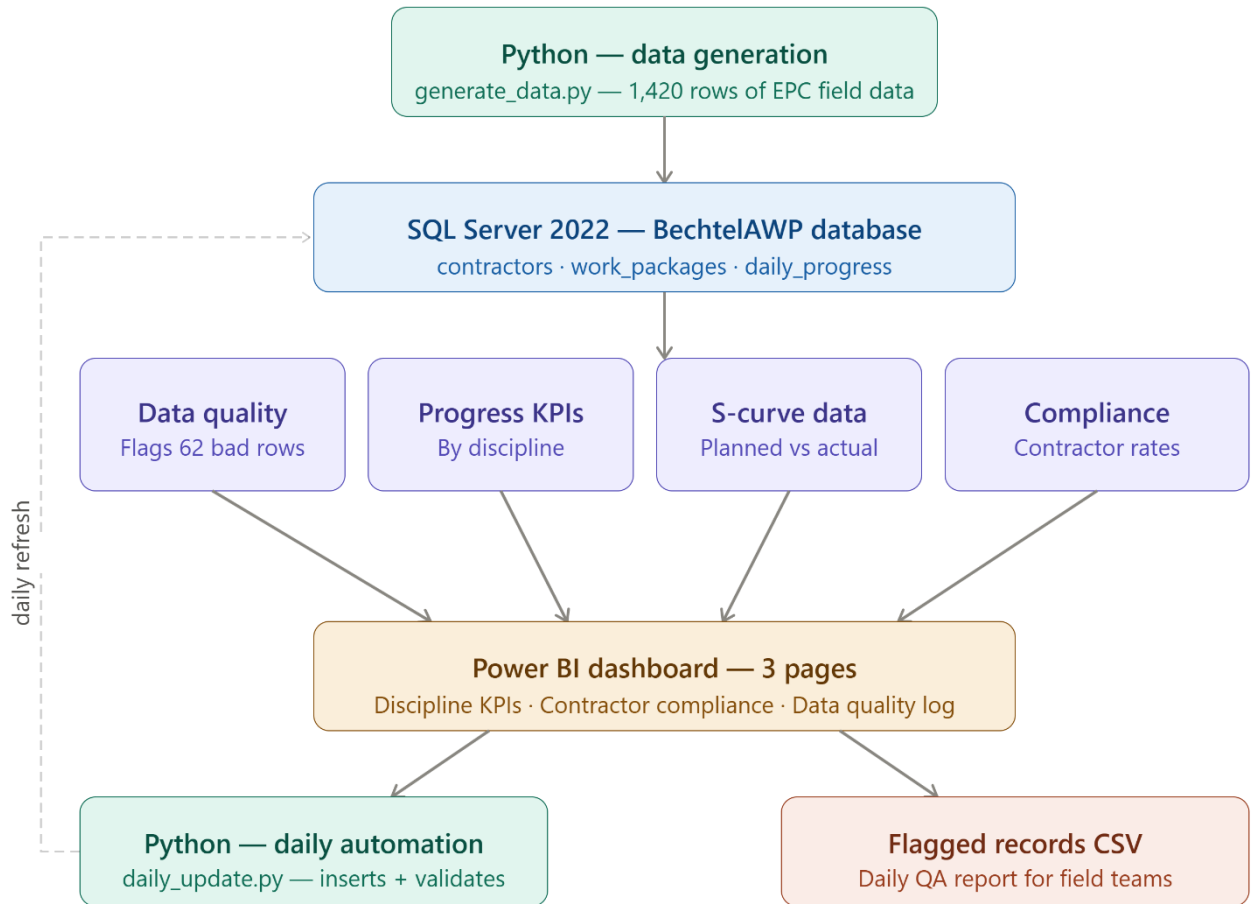
3. Software Requirements

- SQL Server 2022
- SQL Server Management Studio
- Python 3.12
- Power BI Desktop

- VS Code
- ODBC Driver 17 for SQL Server

4. Dashboard overview — 3 pages explained

AWP construction progress dashboard — system architecture



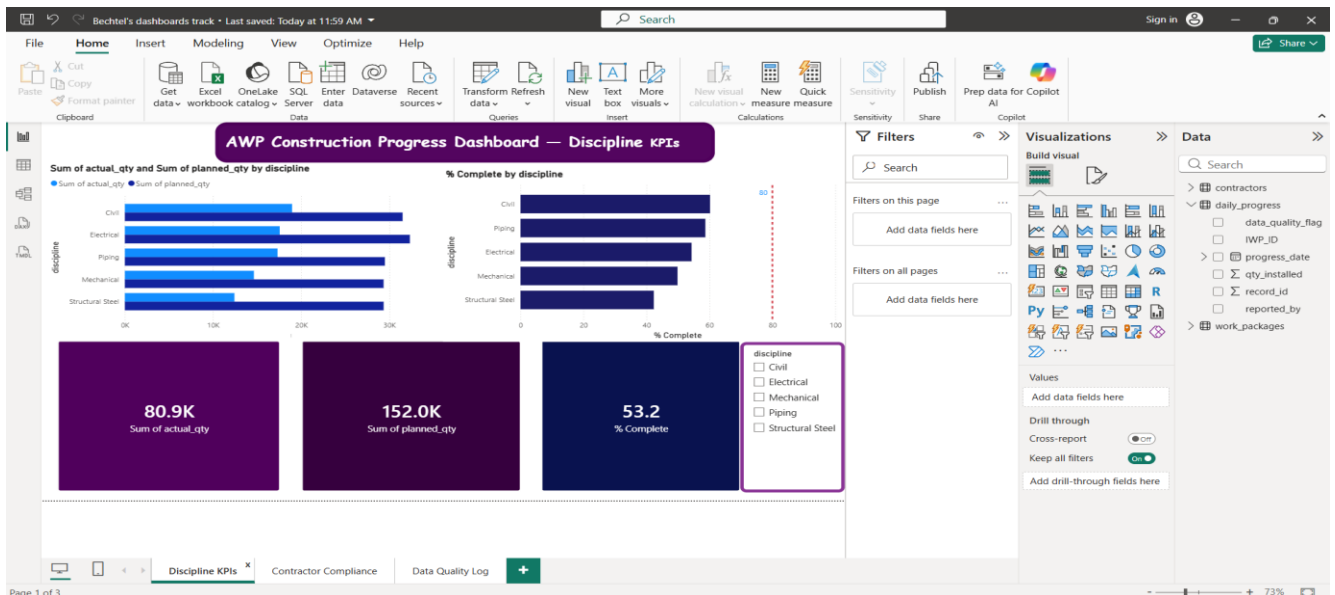
Page 1 — Discipline KPIs

Shows overall construction progress broken down by discipline. Use this page for daily stand-ups and executive briefings.

- **Bar chart (top left)** — actual vs planned quantities per discipline
- **% Complete chart (top right)** — completion rate with 80% target line in red
- **KPI cards** — total actual qty, total planned qty, overall % complete
- **Slicer** — filter all visuals by discipline

NOTE: The red dashed line at 80% is the target threshold. Any discipline bar falling short of this line requires escalation to the field supervisor.

Figure 1: Discipline KPIs

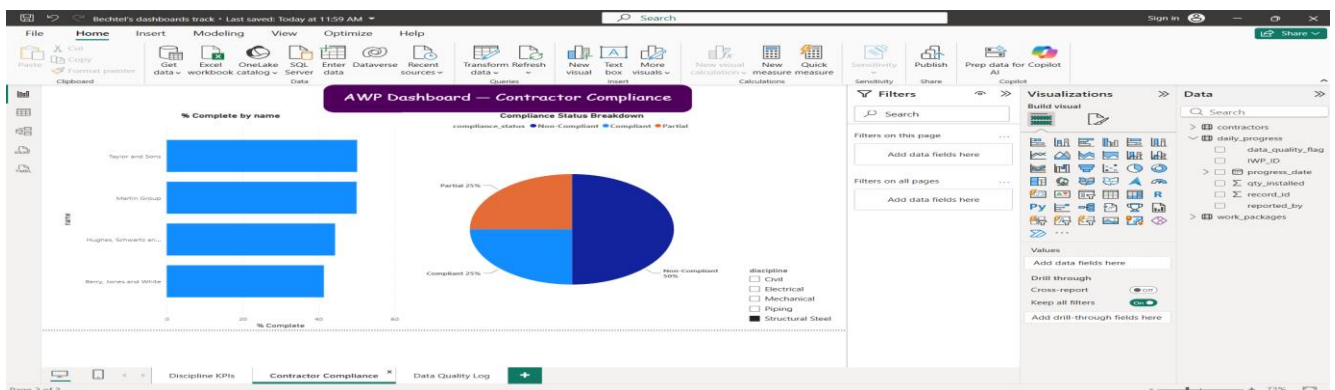


Page 2 — Contractor Compliance

Tracks contractor performance and digital reporting compliance. Use this page for weekly contractor review meetings.

- **Bar chart (left)** — % complete ranked by contractor, highest to lowest
- **Pie chart (right)** — compliance status breakdown (Compliant / Non-Compliant / Partial)
- **Slicer** — filter by discipline to see contractors within a specific trade

Figure 2: Contractor Compliance



Page 3 — Data Quality Log

Flags records that fail validation checks. This page should be reviewed every morning before the daily progress meeting.

- **Left table** — flagged work packages with quality reason codes
- **Right table** — flagged daily progress records (negative qty, missing qty, future dates)

Figure 3: Data Quality Log

5. How to refresh the dashboard

The dashboard connects directly to SQL Server. After new data is loaded, follow these steps to refresh all visuals:

Step 1 Open AWP_Dashboard.pbix in Power BI Desktop

Step 2 Click Home in the top menu bar

- Step 3** Click Refresh in the Queries section
- Step 4** Wait for the data to reload — this typically takes 5-15 seconds
- Step 5** All three pages will update automatically
- Step 6** Review Page 3 (Data Quality Log) first to check for any new flags

WARNING: Do not close Power BI while a refresh is in progress. If the refresh fails, check that SQL Server is running on your machine.

The screenshot shows the Power BI Desktop interface. The top ribbon has the 'Queries' section highlighted, with the 'Refresh' button circled in blue. The main area displays a table titled 'AWP Dashboard — Data Quality Log'. The table has columns: IWP_ID, discipline, status, Sum of planned_qty, Sum of actual_qty, % Complete, and Quality Flag. The table lists various construction items and their completion status. The right-hand pane shows the 'Filters' and 'Visualizations' sections, with the 'Data' section expanded to show a list of fields including contractors, daily_progress, data_quality_flag, IWP_ID, progress_date, record_id, reported_by, and work_packages. The bottom status bar indicates 'Page 3 of 3' and 'Discipline KPIs', 'Contractor Compliance', and 'Data Quality Log' tabs.

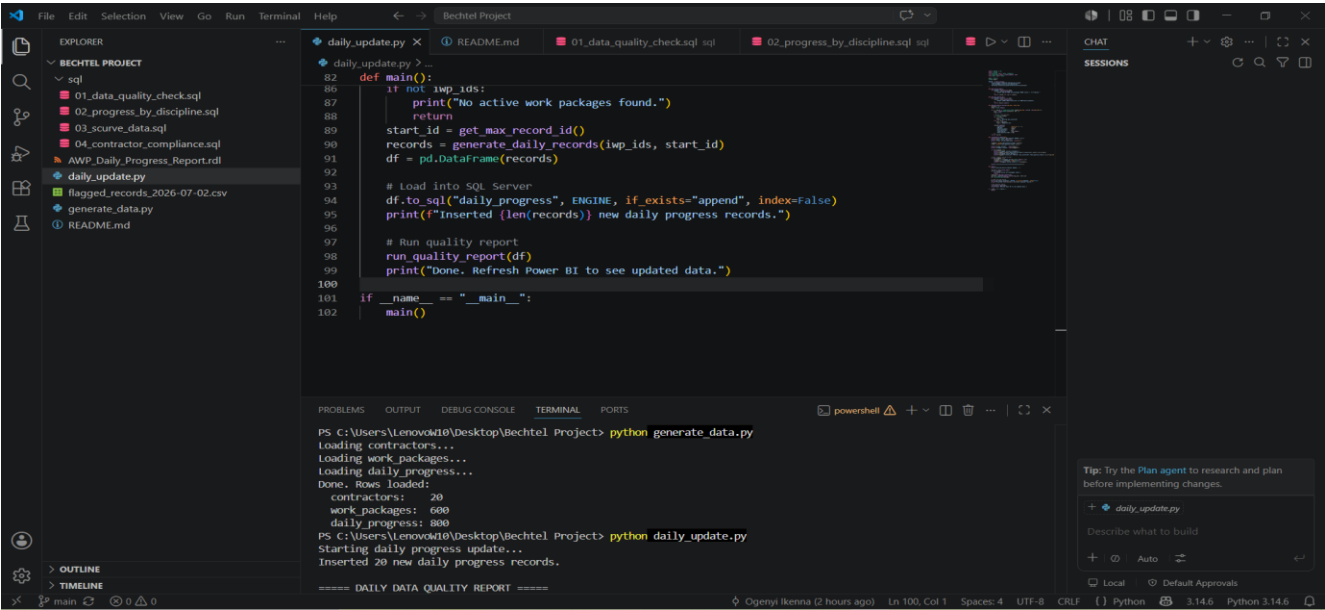
IWP_ID	discipline	status	Sum of planned_qty	Sum of actual_qty	% Complete	Quality Flag
IWP-0006	Structural Steel	Complete	345.50	0.00	0.00	Missing actual start
IWP-0031	Electrical	Complete	256.40	263.50	102.77	Finish before start
IWP-0035	Civil	Complete	234.60	228.90	97.57	Missing planned start
IWP-0039	Piping	Complete	113.20	83.70	73.94	Finish before start
IWP-0066	Structural Steel	Complete	143.10	30.50	21.31	Finish before start
IWP-0082	Piping	Complete	483.10	329.00	68.10	Finish before start
IWP-0094	Structural Steel	Complete	322.60	171.20	53.07	Finish before start
IWP-0103	Structural Steel	Complete	218.30	223.50	102.38	Finish before start
IWP-0149	Civil	Complete	143.90	0.00	0.00	Missing actual start
IWP-0165	Civil	Complete	98.40	91.20	92.68	Finish before start
IWP-0166	Electrical	Complete	44.10	41.10	93.20	Finish before start
IWP-0180	Structural Steel	Complete	476.10	460.50	96.72	Missing planned start
IWP-0207	Piping	Complete	272.70	257.30	94.35	Missing planned start
IWP-0209	Piping	Complete	276.60	351.10	126.93	Over-reported quantity
IWP-0220	Electrical	Complete	188.00	193.10	102.71	Finish before start
IWP-0236	Piping	Complete	85.60	87.90	102.69	Finish before start
IWP-0256	Electrical	Complete	82.40	76.50	92.84	Finish before start
IWP-0270	Structural Steel	Complete	468.90	246.60	52.59	Finish before start
IWP-0283	Civil	Complete	343.30	591.90	172.41	Over-reported quantity
IWP-0293	Piping	Complete	26.50	26.80	101.13	Missing planned start
IWP-0294	Structural Steel	Complete	274.50	0.00	0.00	Missing actual start
IWP-0380	Electrical	Complete	48.20	58.40	121.16	Over-reported quantity
IWP-0394	Civil	Complete	129.80	72.20	55.62	Finish before start
IWP-0446	Civil	Complete	59.60	0.00	0.00	Missing actual start
IWP-0458	Electrical	Complete	108.90	103.20	94.77	Finish before start
Total			6,506.10	5,225.10	80.31	

6. Running the daily automation script

The daily_update.py script simulates what Bechtel's Construction Automation team runs every morning. It inserts new daily progress records into the database, runs data quality checks, and exports a flagged records CSV report.

How to run it

Step 1	Open VS Code and navigate to the Bechtel Project folder
Step 2	Open the terminal (Terminal menu → New Terminal)
Step 3	Run the script by typing: python daily_update.py
Step 4	The terminal will print a data quality report showing clean vs flagged records
Step 5	A CSV file named flagged_records_YYYY-MM-DD.csv is saved to the project folder
Step 6	Refresh Power BI to see the new records appear on Page 3



Expected terminal output

```
Starting daily progress update...

Inserted 20 new daily progress records.

===== DAILY DATA QUALITY REPORT =====
Date: 2026-07-02
Total records generated: 20
Clean records: 17
Flagged records: 3

Flagged records breakdown:
Missing qty_installed 1
Negative qty 2

Flagged records saved to: flagged_records_2026-07-02.csv

=====

Done.Refresh Power BI to see updated data.
```

7. Interpreting data quality flags

The system automatically detects and flags the following data quality issues. Flagged records appear on Dashboard Page 3 and in the daily CSV report.

Work package flags (left table on Page 3)

Flag	Meaning	Action required
Over-reported quantity	Actual qty > 110% of planned	Verify with site supervisor
Finish before start	Actual finish < actual start date	Correct dates in source system
Marked complete, future date	Status = Complete but planned finish is in the future	Update planned finish date
Missing planned start	Planned start date is NULL	Enter planned start in system
Missing actual start	In Progress/Complete but no actual start date recorded	Enter actual start date

Daily progress flags (right table on Page 3)

Flag	Meaning	Action required
Negative qty	qty_installed is negative	Re-enter correct quantity
Missing qty_installed	qty_installed is NULL	Field reporter to resubmit
Future date	progress_date is in the future	Correct the date — likely a typo

8. Common troubleshooting

Power BI shows no data after refresh

Check that SQL Server is running. Open SSMS and verify you can connect to DESKTOP-FKU7PGG. If the service is stopped, open Services (Windows search) and start SQL Server (MSSQLSERVER).

Python script fails with connection error

Verify the connection string in `daily_update.py` matches your server name. Open a terminal and test: `python -c "from sqlalchemy import create_engine; print('OK')"` — if this fails, reinstall pyodbc.

`daily_update.py` runs but inserts 0 records

This means there are no work packages with status = In Progress. Run the `generate_data.py` script again to repopulate the database, or manually update some rows in SSMS.

Power BI cannot connect to SQL Server

In the Get Data dialog, make sure Encrypt is set to Optional and Trust Server Certificate is ticked. The server name must be DESKTOP-FKU7PGG exactly — check for typos.

Data Quality Log shows no flagged rows

The filter on the `data_quality_flag` column may have been reset. Click the right table, go to Filters, expand `data_quality_flag`, and tick Future date, Missing qty_installed, and Negative qty.

9. Version History

Version	Date	Changes
1.0	July 2026	Initial release

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